PROTOCOL TRUTH

Hypurrliquid Yellow Paper

The yellow paper tracks the closest thing this repo has to a technical specification: begin-block ordering, exchange state, action surfaces, hashing, routing, liquidation, and which claims are confirmed versus merely observed or inferred.

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9 min
~1,972 words

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The protocol-truth reference for the Hyperliquid L1 reimplementation. Every claim here is tagged with its evidence level.

Evidence levels: `CONFIRMED` = verified from binary RE or official docs. `OBSERVED` = seen in runtime/snapshots. `IMPLEMENTED` = coded in hlx. `INFERRED` = deduced from behavior.

Use the White Paper for narrative architecture. Use the Hypurrliquid Paper for the full RE reference with Mermaid diagrams. Use the Action Inventory for the current action-family and sub-variant tables. Use the Truth Register when you want the compact fact ledger instead of the full chapter narrative. For the denser subsystem splits, use Hashing, Staking & Validators, Bridge2, and Outcomes.

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0. Truth Surfaces

This paper is the technical narrative, not the only truth surface. Read it together with:

- Truth Register for the compact confirmed/active fact ledger
- Open Claims for unresolved work
- Protocol Scope Matrix for `protocol` vs `testnet_impl` vs `local_impl`
- Action Inventory for action-family and sub-variant tables

1. Block Lifecycle (46 Hook Points)

`CONFIRMED` from binary VA 0x01e748e0 and string at 0x539f8a.

1.1 Pipeline

```
RecoverUsers → BeginBlock (9+5 effects) → DeliverSignedActions (97 type
→ ProcessFills → EVM Block → EndBlock → Commit → VoteAppHash
```

1.2 BeginBlock – 9 Named Effects

#	Effect	Guard Interval	Evidence
1	`update_oracle`	–	`CONFIRMED`
2	`distribute_funding`	8000ms	`CONFIRMED`
3	`apply_bole_liquidations`	–	`CONFIRMED`
4	`update_funding_rates`	8000ms	`CONFIRMED`
5	`refresh_hip3_stale_mark_pxs`	from snapshot	`CONFIRMED`
6	`prune_book_empty_user_states`	60000ms	`CONFIRMED`
7	`update_staking_rewards`	–	`CONFIRMED`
8	`update_action_delayer`	–	`CONFIRMED`
9	`update_aligned_quote_token`	–	`CONFIRMED`

1.3 Supplementary Effects

Effect	Guard	Evidence
`check_trigger_orders`	–	`IMPLEMENTED`
`cancel_aggressive_orders_at_oi_cap`	1000ms	`CONFIRMED`
`validator_l1_vote_tracker_prune_expired`	60000ms	`CONFIRMED`
`reset_recent_ois`	1000ms	`CONFIRMED`
`update_stale_mark_guards`	per-book	`CONFIRMED`

Current local `main` matches the widened 9-effect note on this ordering. What remains open on ActionDelayer is control semantics (`enabled`, `delayer_mode`, `status_guard`, `vty_trackers`), not the named drain slot.

1.4 EndBlock

Mark prices updated from book mid-prices: `mark_prices[asset] = book.mid_price()`.

1.5 VoteAppHash

Every 2000 blocks. Quorum at 2/3+ stake. Final `app_hash` is `first_16(SHA-256(L1_combined)) || first_16(SHA-256(EVM_combined))`, where each half comes from the `rmp_serde → blake3 XOF (2048B) → paddw u16` LtHash pipeline rather than one flat 14-category digest.

1.6 Consensus / Topology Surface

Current confirmed testnet-facing networking and consensus facts:

- ingress is concentrated through a broadcaster layer
- proposer rotation uses stake-weighted `RoundRobinTtl`
- QC and TC certificates drive the two-chain commit rule

- validator/sentry admission checks gate the transport layer
- ports `4001` and `4002` split block-streaming and peer/RPC verification traffic

Treat the exact operator deployment shape as still chain-scoped, but keep these surfaces aligned across lifecycle, gossip, and consensus notes.

1.7 Validator Lifecycle / Jail Surface

Current confirmed validator-control facts:

- epochs are time-based via `epoch_duration_seconds`
- active validators are recomputed at epoch boundaries
- signer admission checks search `epoch_states[cur_epoch]`
- jailed signers are excluded from normal proposer / signing flow
- `CSignerAction` currently has `unjailSelf`, `jailSelf`, and `jailSelfIsolated`
- manual unjail is rate-limited by `signer_to_last_manual_unjail_time`

What remains open on this lane is threshold/trigger detail, not the existence of the lifecycle surface.

---

2. Exchange State (57 Fields)

`CONFIRMED` from serde struct chain at `.rodata` offset `0x543428` in testnet binary `331bef9b` (2026-04-03).

#	Field	Type	Parsed	Evide
0	`locus`	Locus(15)	Yes	`CONF`
1	`perp_dexs`	Vec\<PerpDex>	Yes	`CONF`
2	`spot_books`	Vec\<SpotBook>	Yes	`CONF`
3	`agent_tracker`	AgentTracker(5)	Yes	`CONF`
4	`funding_distribute_guard`	BucketGuard	Yes	`CONF`
5	`funding_update_guard`	BucketGuard	Yes	`CONF`
6	`sub_account_tracker`	SubAccountTracker(2)	Yes	`CONF`
7	`allowed_liquidators`	Vec\<Address>	Yes	`CONF`
8	`bridge2`	Bridge2(10)	Yes	`CONF`
9	`staking`	Staking(6)	Yes	`CONF`
10	`c_staking`	CStaking(23)	Yes	`CONF`
11	`funding_err_dur_guard`	BucketGuard	Yes	`CONF`
12	`max_order_distance_from_anchor`	f64	Yes	`CONF`
13	`scheduled_freeze_height`	u64	Yes	`CONF`
14	`simulate_crash_height`	u64	Yes	`CONF`
15	`validator_power_updates`	Vec	Skip	`CONF`
16	`cancel_oi_cap_guard`	BucketGuard	Yes	`CONF`
17	`last_hlp_cancel_time`	f64	Yes	`CONF`
18	`post_only_until_time`	f64	Yes	`CONF`
19	`post_only_until_height`	u64	Yes	`CONF`
20	`spot_twap_tracker`	JSON	Yes	`CONF`
21	`user_to_display_name`	HashMap	Yes	`CONF`

#	Field	Type	Parsed	Evide
22	`book_empty_user_states_pruning_guard`	BucketGuard	Yes	`CONF`
23	`user_to_scheduled_cancel`	HashMap	Yes	`CONF`
24	`hyperliquidity`	2 fields	Yes	`CONF`
25	`spot_disabled`	bool	Yes	`CONF`
26	`prune_agent_idx`	u64	Yes	`CONF`
27	`max_hlp_withdraw_fraction_per_day`	f64	Yes	`CONF`
28	`hlp_start_of_day_account_value`	f64	Yes	`CONF`
29	`register_token_gas_auction`	GasAuction(4)	Yes	`CONF`
30	`perp_deploy_gas_auction`	GasAuction(4)	Yes	`CONF`
31	`spot_pair_deploy_gas_auction`	GasAuction(4)	Yes	`CONF`
32	`hyper_evm`	HyperEvmState(13)	Yes	`CONF`
33	`vtg`	JSON	Yes	`CONF`
34	`app_hash_vote_tracker`	3 fields	Yes	`CONF`
35	`begin_block_logic_guard`	BucketGuard	Yes	`CONF`
36	`user_to_evm_state`	HashMap (778K)	Skip	`CONF`
37	`multi_sig_tracker`	1 field	Yes	`CONF`
38	`reserve_accumulators`	JSON	Yes	`CONF`
39	`partial_liquidation_cooldown`	f64 (30.0)	Yes	`CONF`
40	`evm_enabled`	bool	Yes	`CONF`
41	`validator_l1_vote_tracker`	3 fields	Yes	`CONF`
42	`staking_link_tracker`	JSON	Yes	`CONF`
43	`action_delayer`	ActionDelayer(9)	Yes	`CONF`

#	Field	Type	Parsed	Evide
44	`default_hip3_limits`	JSON	Yes	`CONF`
45	`hip3_stale_mark_px_guard`	BucketGuard	Yes	`CONF`
46	`disabled_precompiles`	Vec	Yes	`CONF`
47	`lv`	JSON	Yes	`CONF`
48	`hip3_no_cross`	bool	Yes	`CONF`
49	`initial_usdc_evm_system_balance`	f64	Yes	`CONF`
50	`validator_l1_stream_tracker`	JSON	Yes	`CONF`
51	`last_aligned_quote_token_sample_time`	f64	Yes	`CONF`
52	`user_to_evm_to_l1_wei_remaining`	HashMap	Yes	`CONF`
53	`dex_abstraction_enabled`	bool	Yes	`CONF`
54	`hilo`	HyperliquidityImpactLimiter(3)	Yes	`CONF`
55	`oh`	Vec\<OracleSnapshot>(100)	Yes	`CONF`
56	`hpt`	AlignedQuoteTokenTracker	Yes	`DISC`

57/57 fields declared. 55/57 deeply parsed from snapshots. 2 skipped (user_to_evm_state, validator_power_updates).

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3. Action Surface (97 Variants, 126 Sub-Types)

`CONFIRMED` from binary dispatcher at VA 0x0272a320. 90 mainnet + 7 testnet.

Category	Count	Key Types
Trading	9	order, cancel, cancelByCloid, modify, batchModify, TWAP, liquidate
Margin	3	updateLeverage, updateIsolatedMargin, topUpIsolatedOnlyMargin
Transfers	6	usdSend, spotSend, sendAsset, withdraw3, sendToEvmWithData
Sub-Accounts	3	createSubAccount, subAccountTransfer, subAccountSpotTransfer
Vaults	6	createVault, vaultTransfer, vaultDistribute, vaultModify, netChildVaultPositions, setVaultDisplayName
Staking	5	tokenDelegate, claimRewards, linkStakingUser, registerValidator, extendLongTermStaking
Account	7	approveAgent, setDisplayName, setReferrer, convertToMultiSigUser, approveBuilderFee, startFeeTrial, userPortfolioMargin
DeFi	5	borrowLend, userDexAbstraction, userSetAbstraction, agentSetAbstraction, agentEnableDexAbstraction
Governance	3	govPropose, govVote, voteGlobal (22 sub-types)
HIP-3/4	3	SetGlobal (6 sub), perpDeploy (8 sub), spotDeploy
Validator	9	CValidator, CSigner, validatorL1Vote, validatorL1Stream, bridge votes (5)
System	11	SystemBole, SystemSpotSend, CWithdraw, CUserModify, EvmUserModify, etc.
Special	7	evmRawTx, multiSig, noop, VoteAppHash, ForceIncreaseEpoch, userOutcome (4 sub), settleFraction
Additional	8	priorityBid, registerSpot, disableDex, cancelAllOrders, quarantineUser, etc.

Current testnet notes widen the action surface to 97 variants. The local engine has broad typed coverage, but parity work remains active and this page should not be read as a blanket “zero gaps” claim.

4. Funding Rate

`CONFIRMED` from binary assertions.

- **EMA**: DeterministicEma with fraction-based accumulation (num/denom). `assert!(decay <= 1.)`, `assert!(self.denom > 0.)`.
- **Funding payment**: `payment = szi × oracle_px × (cum_funding_global − pos.cum_funding_entry)`
- **Direction**: Positive rate = longs pay shorts.
- **Guard interval**: 8000ms for both distribute and update.
- **Per-asset multiplier**: Set via `SetFundingMultiplier` VoteGlobal.

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5. Liquidation & ADL

`CONFIRMED` from binary constants and docs.

- **Margin check**: `margin_available = account_value − initial_margin_required`. Liquidatable when < 0.
- **Isolated**: Per-position check with allocated margin.
- **BOLE liquidation threshold**: Health factor < 0.8.
- **BOLE cooldown**: 30 seconds for partial liquidations.
- **BOLE types**: Partial (50% seize), Market (full repay), Backstop (default liquidator).
- **ADL**: Triggered when shortfall persists after liquidation. Counterparties ranked by ROE (highest first). Position reduced proportionally.
- **ADL priority**: `ROE = unrealized_pnl / margin_allocated`.

6. State Hashing

`CONFIRMED` algorithm: `rmp_serde → blake3 XOF (2048B) → paddw u16 (LtHash16) → SHA-256 (32B)`.

- **11 L1 categories** + **3 EVM categories** = 14 accumulators.
- **L1 accumulators**: Running sums from genesis maintained in live state, not the stale BSS cache that earlier notes treated as canonical.
- **EVM accumulators**: accounts_hash, contracts_hash, storage_hash.
- **Heartbeat** `concise_lt_hashes`: EVM only. The traced heartbeat surface does not carry the full L1 accumulator state.

6.1 AppHash Split

Current repo truth is that the final `app_hash` is `first_16(SHA-256(L1_combined)) || first_16(SHA-256(EVM_combined))`, not a single SHA-256 over one merged 14-accumulator buffer.

6.2 RespHash Backend Split

Current repo truth is intentionally narrower than "hashing is solved":

- the code explicitly tracks `Mainnet` versus `Testnet` RespHash backends
- the backend split is real and should remain chain-scoped in docs and replay work
- the local implementation does **not** yet claim full mainnet serializer parity across all response families

Treat `RespHashMode::Mainnet` as backend selection, not as proof that every mainnet response serializer is already closed.

7. CoreWriter (EVM → L1)

`CONFIRMED` from binary and official docs.

- **Precompile address**: `0x3333333333333333333333333333333333333333`
- **Function**: `sendRawAction(bytes)` selector `0x17938e13`
- **Encoding**: `abi.encodePacked(uint8(1), uint24(actionId), abi.encode(params...))`
- **15 action types**: LimitOrder, VaultTransfer, TokenDelegate, StakingDeposit, StakingWithdraw, SpotSend, UsdClassTransfer, FinalizeEvmContract, AddApiWallet, CancelByOid, CancelByCloid, ApproveBuilderFee, SendAsset, ReflectEvmSupplyChange, BorrowLend.
- **Delay**: `delay_scale × 3000ms` (configurable). Actions fire in next block's begin_block Effect 8.
- **In replay mode**: Raw tx calldata scanned via RLP parser without EVM execution.

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8. Bridge Finalization and Validator-Set Updates

`CONFIRMED` from Bridge2 state naming, bridge action inventory, and the current control-plane notes.

- **Bridge2 state** includes `withdrawal_signatures`, `withdrawal_finalized_votes`, `finished_withdrawal_to_time`, `validator_set_signatures`, `validator_set_finalized_votes`, `bal`, `last_pruned_deposit_block_number`, and `oaw` (`bool`).
- **Withdrawal entrypoint**: `withdraw3`.
- **Bridge signing action**: `ValidatorSignWithdrawal`.
- **Finalize actions**: `VoteEthFinalizedWithdrawal` and `VoteEthFinalizedValidatorSetUpdate`.

- **Validator-set update signing**: `SignValidatorSetUpdate`.
- **Signer model**: current repo truth keeps bridge signing on validator signer keys, not an unrelated bridge-only signer family.

What remains open is exact dispute-period closure and some Ethereum-side operational details, not whether bridge finalization is a first-class staged state machine.

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9. Aligned Quote Token (HPT)

`DISCOVERED` from Hyperliquid docs + API.

- **Exchange field [56]** `hpt` = AlignedQuoteTokenTracker.
- **Per-token state**: isAligned, firstAlignedTime, evmMintedSupply, dailyAmountOwed, predictedRate.
- **Validator sidecar**: `aqa-publisher` (github.com/native-markets/aqa-publisher).
- **Rate source**: 30-day SOFR average from NY Fed, FRED, OFR (median of 3 sources).
- **Submission**: `validatorL1Stream` action with `riskFreeRate` field, daily at 22:00 UTC.
- **Protocol rate**: Stake-weighted median of all active validator votes.
- **Fee benefits**: 20% lower taker, 50% better maker rebate for aligned stablecoins.

This is also a begin-block surface: `update_aligned_quote_token` is the current effect-9 lane in the widened testnet note.

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10. Outcomes and Settlement Surface

`CONFIRMED` for the baseline market model; `HYPOTHESIS` for the leading solvency-risk lane.

- **Risk mode**: 1x isolated-only, no leverage, no funding.
- **Settlement authority**: `oracleUpdater` posts final outcome via `SettleOutcome`.
- **Post-settlement behavior**: open orders auto-cancel via `outcomeSettledCanceled`; new orders reject via `outcomeSettledRejected`.
- **Question metadata**: `fallbackOutcome`, `namedOutcomes`, `settledNamedOutcomes`.
- **User token operations**: `SplitOutcome`, `MergeOutcome`, `MergeQuestion`, `NegateOutcome`.

Current open risk lane:

- `MergeQuestion` / fallback reconciliation is still the leading higher-order solvency hypothesis.
- The repo should treat question-level accounting as riskier than the simpler YES/NO pair mechanics until the binary path is closed.

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11. Gossip Protocol

`CONFIRMED` from binary RE.

- **Frame format**: `[u32 BE len][u8 kind][body]`. Traced data frames often use LZ4 with `kind=0x01`, while greetings use `kind=0x00`.
- **Port 4001 greeting**: bincode-fork `TcpGreeting` with chain-based variant index. Current live observations are testnet variant `3` (8 bytes total) and mainnet variant `2` (7 bytes total).
- **Broader chain enum**: the ABCI-stream chain mapping still tracks `Local=0`, `Sandbox=1`, `Testnet=2`, `Mainnet=3`; do not conflate that with the observed 4001 `TcpGreeting` variant index.
- **Port 4003 authenticated greeting**: `keccak256("Hyperliquid Consensus Payload" + 0x00 + bincode(content))`; content is bincode, not msgpack.

- **Connection**: Server-side `connection_checks` — unknown peers rejected after 5s timeout.
- **Loopback**: Dominated by `MsgConcise` / `OutMsgConcise` (tag 27/28).

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12. Guard System

All rate-limited effects use `BucketGuard`:

```
struct BucketGuard {
    last_time: Option<u64>,   // ms
    interval_ms: u64,
}
```

Guard	Interval	Effect
`funding_distribute_guard`	8000ms	Funding settlement
`funding_update_guard`	8000ms	Premium sampling
`cancel_oi_cap_guard`	1000ms	OI cap order cancellation
`book_empty_user_states_pruning_guard`	60000ms	Book cleanup
`reset_recent_oi_guard`	1000ms	HIP-3 OI reset
`hip3_stale_mark_px_guard`	From snapshot	Stale mark refresh
`begin_block_logic_guard`	1000ms	Reentrancy protection
`funding_err_dur_guard`	60000ms	Funding error duration

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13. Implementation Status

Component	Status	Tests
Action handlers (97)	All implemented	330+
Begin-block (9 + 5)	All implemented	Tested
Bootstrap (57 fields)	55/57 parsed	Bootstrap test
CoreWriter decoder	15 actions + RLP scanner	7 tests
ADL + BOLE liquidation	Execution implemented	ADL plan + candidates tests
Pipeline stages	6 stages with real logic	4 tests
RPC (30+ endpoints)	Consolidated in hl-rpc	Info tests
Integration tests	18 E2E devnet tests	All pass
Total	**482 tests, 0 failures**	**Release: 11MB**